



5110 - omega Centauri: the coolest pieces of its population puzzle

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) NGC-5139
	2		NIRCam Imaging	(1) NGC-5139

ABSTRACT

As the largest and oldest coeval stellar populations, globular clusters (GCs) trace the origin of chemical abundances in the Galaxy and offer unique insight into the properties of ancient metal-poor stars. The long-held notion of chemical homogeneity among cluster members has been firmly ruled

out by two decades of space-based photometry that revealed complex color-magnitude diagrams (CMDs), displaying multiple populations (mPOPs) of stars with distinct compositions. The enrichment mechanism responsible for this phenomenon remains uncertain, in part, because most GC studies have been restricted to bright main sequence or giant stars.

We propose to use JWST in direct-imaging mode with NIRCам and NIRISS to obtain high-precision photometry and astrometry of the faintest objects in omega Centauri (NGC 5139) – the most massive GC that may host as many as 15 sub-populations. We intend to explore the transition in the CMD between stars fusing Hydrogen and non-fusing brown dwarfs (BDs). Observations of BDs will provide precise ages and compositions for individual sub-populations as well as probe the hypothetical dependence of abundances on the stellar mass, allowing direct tests of various mPOPs formation theories. The new data will be also used for calibrating metal-poor models of BDs, thus serving as a 'Rosetta-stone' between the poorly understood physics of low-temperature atmospheres and reliable measurements of chemical abundances in higher-mass stars.

In addition, the proposed program will be sensitive to the entire white dwarf (WD) cooling sequence, allowing the search for IR excess in WD colors as a signature of ancient planetary systems.

OBSERVING DESCRIPTION

In this program, we propose to use JWST in direct-imaging mode with NIRCам (and in parallel with NIRISS to obtain high-precision photometry and astrometry of the faintest objects within the massive globular cluster (GC): omega Centauri (NGC 5139).

We need to map a relatively large field of view to have a statistically significant number of expected white dwarfs (WDs), low mass Main-Sequence stars, and brown dwarfs (BDs), therefore we use NIRCам and in parallel NIRISS (the widest imagers on the JWST focal plane) as coordinated parallel observations.

As GC's members are completely lost in the mixture of objects in their foreground and background, astrometry is a fundamental part of this investigation, and we intend to probe both the astrometric and photometric capabilities of JWST in this program. Multiple dithered images each with relatively high S/N are necessary to provide a number of inter-comparisons of detectors' regions to calibrate (or to correct for residual in) the geometric distortion of the camera, the spatial and temporal variability of the PSFs, and the local photometric zero points.

We will take advantage as much as possible of existing archival HST/ACS/WFC/F814W deep images (from GO-14118 and GO-14662) to have a first epoch enable to separate members stars from field objects in background and foreground of the target GC, however only the WDCS were

JWST Proposal 5110 (Created: Thursday, April 9, 2026, 12:00:09PM Eastern Standard Time) - Overview

observed completely. We will also use, at their full potential, the IR images available in WFC3/IR/F110W + F160W collected as parallel fields under the same HST programs, although much shallower than the IR JWST observations here proposed, but nevertheless the space-based deepest IR images available.

To get some proper motion information for BDs, which are undetected in HST archival data, we have split our observations in two halves that will be taken about 1 year apart. One in F322W2 and one in F444W. Not only this provide a color information for the coolest BDs. but also distinguishes them from the much faster local BDs in the Galactic field, and with respect to stationary background far away galaxies.

Precise and accurate astrometry, will not only be used for proper-motion cluster memberships, but it will be used also for analyses of the GC internal kinematics (such investigations are beyond Gaia's capabilities, as the stars too red & faint and in crowded environments).

We have chosen the JWST NIRCAM ultra-wide in two filter-pairs: (F150W2,F322W2) and (F090W,F444W), which can be used for simultaneous observations. The first pair for the large throughput suitable for deep observations of old cool faint WDs and BDs; the second pair to enable highest angular resolution and astrometric precision (F090W) of all MS stars and WDs and to get a second epoch for BDs (F444W). We also plan to have NIRISS parallel images in two filters and with a time baseline of ~1yr to increase the field coverage.

Proposal 5110 - Targets - omega Centauri: the coolest pieces of its population puzzle

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-5139	RA: 13 25 27.4200 (201.3642500d)	Proper Motion RA: -3.24 mas/yr	
			Dec: -47 35 39.10 (-47.59419d)	Proper Motion Dec: -6.729999904564465 mas/yr	
			Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				

Proposal 5110 - Observation 1 - omega Centauri: the coolest pieces of its population puzzle

Thu Apr 09 17:00:09 GMT 2026

Observation	Proposal 5110, Observation 1										
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Coordinated Parallel Template(s): NIRISS Imaging										
Diagnostics	(Observation 1) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(1)	NGC-5139	RA: 13 25 27.4200 (201.3642500d)			Proper Motion RA: -3.24 mas/yr					
			Dec: -47 35 39.10 (-47.59419d)			Proper Motion Dec: -6.729999904564465 mas/yr					
			Equinox: J2000			Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Stellar Cluster										
	Description=[Globular star clusters]										
Template	NIRCcam Imaging					NIRISS Imaging					
	Module: ALL										
	Subarray: FULL										
	Target Placement: Module gap (large extended source)										
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
	1	FULLBOX		6				1		3-POINT-LARGE-WITH-NIRISS	NO_DITHERING
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	MEDIUM8	9	1	18	18	17007.044		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F140M		NIS	22	1	18	18	17200.306		

Proposal 5110 - Observation 1 - omega Centauri: the coolest pieces of its population puzzle

Special Requirements	Before Date 01-OCT-2024 Aperture PA Range 123.92542306 to 123.92542306 Degrees (V3 124.0 to 124.0) No Parallel Attachments
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Proposal 5110 - Observation 2 - omega Centauri: the coolest pieces of its population puzzle

Thu Apr 09 17:00:09 GMT 2026

Observation	Proposal 5110, Observation 2										
	Diagnostic Status: Warning										
	Observing Template: NIRCам Imaging										
	Coordinated Parallel Template(s): NIRISS Imaging										
Diagnostics	(Observation 2) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(1)	NGC-5139	RA: 13 25 27.4200 (201.3642500d) Dec: -47 35 39.10 (-47.59419d) Equinox: J2000			Proper Motion RA: -3.24 mas/yr Proper Motion Dec: -6.729999904564465 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Stellar Cluster Description=[Globular star clusters]										
Template	NIRCам Imaging					NIRISS Imaging					
	Module: ALL										
	Subarray: FULL										
	Target Placement: Module gap (large extended source)										
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
	1	FULLBOX		6				1		3-POINT-LARGE-WITH-NIRISS	NO_DITHERING
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F444W	MEDIUM8	9	1	18	18	17007.044		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F200W		NIS	22	1	18	18	17200.306		

Proposal 5110 - Observation 2 - omega Centauri: the coolest pieces of its population puzzle

Special Requirements	After Date 01-AUG-2027:00:00:00 Aperture PA Range 123.92542306 to 123.92542306 Degrees (V3 124.0 to 124.0) No Parallel Attachments
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